

Mugume Ronald

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Executive Summary

Enterprise Cloud Architect and hardened Systems Engineer specializing in high-density, multi-tenant Kubernetes infrastructure. Founder of Aerophib Technologies Ltd, dedicated to delivering FAANG-level cloud-native solutions to modern enterprises. Proven track record in replacing fragile legacy setups with immutable GitOps pipelines, implementing zero-trust security paradigms, and ensuring 99.9% uptime for mission-critical stateful workloads.

Core Competencies

- **Container Orchestration:** Kubernetes (Bare-Metal, GKE, EKS, AKS), Docker, Helm, Kustomize
 - **Infrastructure & Automation:** GitOps, Terraform, Terragrunt, HCL, Advanced Bash Scripting
 - **Security & Identity:** Zero-Trust Architecture, Authelia SSO, Cloudflare Tunnels, Network Policies, RBAC
 - **Observability & Telemetry:** Prometheus, Grafana, Loki, Uptime Kuma
 - **Cloud Platforms:** Google Cloud (GCP), AWS, Azure, Hybrid Cloud Deployments
 - **Linux Systems Engineering:** RedHat, Alpine, Arch Linux, Kernel Hardening, VLAN Ingress
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Professional Experience

Founder & Lead Cloud Architect

Aerophib Technologies Ltd | 2023 – Present

- **Enterprise Cloud Transformation:** Spearhead cloud-native consulting engagements, architecting multi-region, highly available infrastructure for SME and enterprise clients to accelerate time-to-market and maximize ROI.
- **Multi-Tenant SaaS Infrastructure:** Engineered a proprietary B2B SaaS routing gateway and bare-metal Kubernetes cluster, enabling dynamic provisioning of isolated client tenants with strictly enforced network boundaries.
- **Zero-Trust Security Implementation:** Architected and deployed comprehensive identity-aware proxy solutions utilizing Authelia and ForwardAuth, securing all internal applications behind military-grade SSO without reliance on expensive proprietary licensing.

- **Stateful Workload Optimization (ERPNext):** Designed and deployed high-availability ERPNext clusters. Resolved critical multi-node NFS mounting bottlenecks by engineering custom Kubernetes volume-affinity scheduling, ensuring stateful workloads perform at peak efficiency with automated disaster recovery.
- **Advanced Observability Pipelines:** Standardized infrastructure telemetry by implementing the Prometheus, Grafana, and Loki stack, providing clients with centralized audit logging and real-time visibility into cluster health and application performance.
- **GitOps & CI/CD Automation:** Replaced manual deployment processes with immutable GitOps pipelines, ensuring infrastructure-as-code (IaC) compliance and executing zero-downtime rollout strategies across production environments.

Open-Source Contributions & Engineering Projects

- **erpnext-kubernetes:** Architected production-ready Helm charts and deployment patterns for ERPNext, specifically addressing local storage provisioning and node affinity for bare-metal clusters.
 - **saas-launchpad:** Developed a custom Python/Flask orchestration engine that dynamically routes and provisions isolated client tenants within a multi-tenant environment.
 - **Infrastructure Provisioning Blueprint:** Created standardized Terraform blueprints to programmatically deploy Google Kubernetes Engine (GKE) clusters with optimized node groups for rapid scaling.
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Education & Certifications

- **Google Cloud Associate Cloud Engineer (ACE)** – Google Cloud
- **Kubernetes & Cloud Native Associate (KCNA)** – Cloud Native Computing Foundation (CNCF)
- **Bachelor of Science in Information Technology**
- *In Progress:* Certified Kubernetes Application Developer (CKAD)